MINOR PROJECT / COMPUTER VISION

Real-Time Sign Language Recognition

Pose + both hands • LSTM sequence model • webcam inference

PROJECT SNAPSHOT

CAMERA-BASED ISOLATED SIGN CLASSIFICATION

Recognize a sign from movement

The system samples a short video, extracts normalized body and hand landmarks, then predicts one label from a fixed vocabulary.

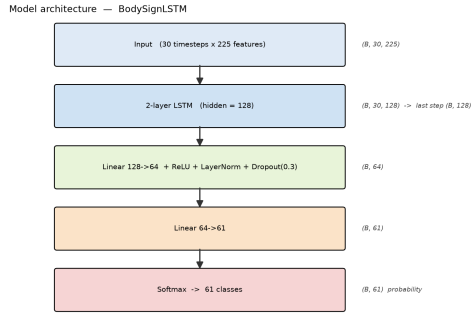
3,630
clips / sequences

61
sign classes

Single-signer dataset; performance on unseen signers is not established.



PROJECT MODEL

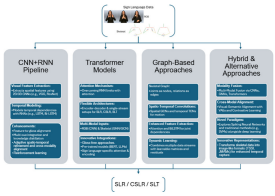


Architecture figure from the project images folder.

TEMPORAL LANDMARK CLASSIFIER

- 33 body landmarks + 21 points per hand
- 30 frames × 225 features per frame
- 2-layer LSTM with a 64-unit classifier
- Mirror augmentation supports handedness variation

RESEARCH CONTEXT / REVIEW PAPER FIG. 1



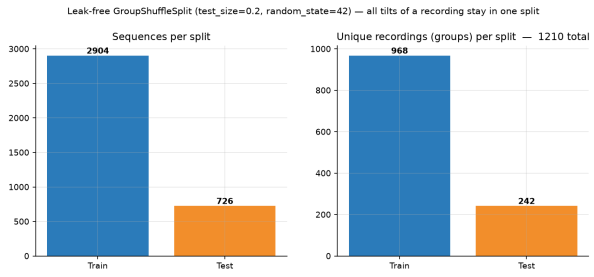
Four method families

CNN–RNN pipelines, Transformers, graph-based models, and hybrid approaches.

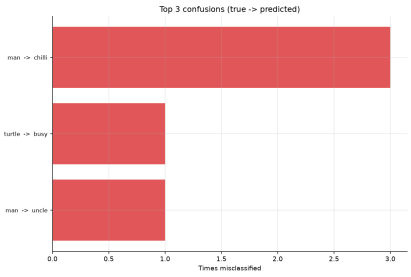
This project follows a pose-based sequence approach. The review notes that landmarks are compact, but occlusion and missing facial cues can limit recognition.

DATA SPLIT & ERROR ANALYSIS

Recordings are grouped before the split so orientation views of one recording stay together.



GROUPED TRAIN / TEST SPLIT



MOST FREQUENT CONFUSION PAIRS

LIMITATION: recordings come from one signer; test performance may not transfer to new people or settings.

RESEARCH SOURCE

Kalinowski, M. & Kostek, B. (2026). Machine learning in sign language: A comprehensive analysis and trend survey. Computer Science Review, 60, 100895. doi:10.1016/j.cosrev.2026.100895.